4. Set Theory

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* Statement or Proposition

All decrialative sentences whose truth value can be determined is called statement or proposition.

* Logical operators

Logical connectives are logical tools which are required to connect ≥ 2 or more statements

Connectives	Symbol	Name
NOT	~ or ¬	Negation
AND	∧	Conjunction
OR	∨	Disjunction
IF.... THEN	→ or ⇒	Implication
IF AND ONLY IF (IFF)	↔ or ⇔	Bi-implication

If p and q are statements then conjunction of p & q is denoted by p ∧ q is a statement which has a truth value T if both p & q are true otherwise it has truth value F.

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Truth value:

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Formula:

$p \leftrightarrow q = (p \rightarrow q) \wedge (q \rightarrow p)$

$T \wedge F = F$

$T \vee F = T$

Que: Construct the truth table for fol

1) $(\sim q \rightarrow \sim p) \vee q$

2) $(p \wedge q) \vee (p \rightarrow r) \vee (\sim q \rightarrow r)$

Sol^n:

p	q	$\sim p$	$\sim q$	$\sim q \rightarrow \sim p$	$(\sim q \rightarrow \sim p) \vee$
T	T	F	F	T	T
T	F	F	T	F	T
F	T	T	F	T	T
F	F	T	T	T	T

~~p q~~

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p	q	r	$\sim q$	$p \wedge q$	$p \rightarrow r$	$\sim q \rightarrow r$
T	T	T	F	T	T	T
T	T	F	F	T	F	T
T	F	T	T	F	T	T
T	F	F	T	F	F	F
F	T	T	F	F	T	T
F	T	F	F	F	T	T
F	F	T	T	F	T	T
F	F	F	T	F	T	F

$(p \wedge q) \vee (p \rightarrow r)$	$(\sim q \rightarrow r) \vee (\sim q \rightarrow r)$
T	T
T	T
T	T
F	F
T	T
T	T
T	T
T	T

3) Show that $p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$ using truth table.

~~p q~~

p	q	r	$(q \vee r)$	$p \wedge (q \vee r)$
T	T	T	T	T
T	T	F	T	T
T	F	T	T	T
T	F	F	F	F
F	T	T	T	F
F	T	F	T	F
F	F	T	T	F
F	F	F	F	F

$(p \wedge q)$	$(p \wedge r)$	$(p \wedge q) \vee (p \wedge r)$
T	T	T
T	F	T
F	T	T
F	F	F
F	F	F
F	F	F
F	F	F
F	F	F

Converse, Inverse & Contrapositive

If p & q are two statements & $p \to q$ is an implication then

1. $q \to p$ is called Converse
2. $\sim p \to \sim q$ is called Inverse
3. $\sim q \to \sim p$ is called Contrapositive

NOTE: Note that every implication is equivalent to its contrapositive

Negation of Compound Statement

i) $\sim(p \wedge q) \equiv \sim p \vee \sim q$

ii) $\sim(p \vee q) \equiv \sim p \wedge \sim q$

iii) $\sim(p \to q) \equiv p \wedge \sim q$

iv) $\sim(p \leftrightarrow q) \equiv (p \wedge \sim q) \vee (q \vee \sim p)$

Que: Write the negation of the following statement

1. If ram is good then shyam is king
2. Either ram is guilty or shyam is not telling truth.

Sol^n: 1. Ram is good $\to p$

Shyam is king $\to q$

$\sim(p \to q) \equiv p \wedge \sim q$

Que:- State converse, inverse & contrapositive
1) If today is Sunday then tomorrow
Monday.
2) If p is a square then q is a re

Ans:- ① Today is Sunday $\rightarrow p$
tomorrow is Monday $\rightarrow q$

$$p \rightarrow q \equiv q \rightarrow p \quad [\text{Converse}]$$
$$\equiv \sim p \rightarrow \sim q \quad [\text{inverse}]$$
$$\equiv \sim q \rightarrow \sim p \quad [\text{Contrapositive}]$$

Write statement also.

Statement pattern
A pattern obtained by combin
two or more statement using logic
connective is called statement patte
following are that the statements patte
① Tautology
In this statement pattern if all
truth values are true then it i
called a tautology
② Contradiction
In this statement pattern if a
truth values are false then it is
called a Contradiction.

③ Contigency
If a statement pattern is
neither a tautology nor a contradiction
then it is called a contigency.

② Preposition like
Ex:- Sun revolves around the earth is always
false and it will be contradiction.

Que:- Check which of the following are
tautology contradiction & contigency.
1) $(p \rightarrow q) \rightarrow (\sim p \vee q)$
$\Rightarrow$ ~~$(\sim p \wedge q)(p \rightarrow q) = \sim p \wedge q$~~

p	q	$p \rightarrow q$	$\sim p$	$\sim p \vee q$	$(p \rightarrow q) \rightarrow (\sim p \vee q)$
T	T	T	F	T	T
T	F	F	F	F	T
F	T	T	T	T	T
F	F	T	T	T	T

∴ It is Tautology.

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2) $((p \vee q) \wedge r) \vee (p \vee q)$

p	q	r	$p \vee q$	$(p \vee q) \wedge r$	$((p \vee q) \wedge r) \vee (p \vee q)$
T	T	T	T	T	T
T	T	F	T	F	T
T	F	T	T	T	T
T	F	F	T	F	T
F	T	T	T	T	T
F	T	F	T	F	T
F	F	T	F	F	F
F	F	F	F	F	F

∴ It is contigency

3) $p \wedge (q \wedge \sim p)$

p	q	$q \wedge \sim p$	$q \wedge \sim p$	$p \wedge (q \wedge \sim p)$
T	T	F	F	F
T	F	F	F	F
F	T	T	T	F
F	F	T	F	F

∴ It is a contradiction

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Show that
$[p \to (q \to r)] \leftrightarrow [p \to (\sim q \vee r)]$

Show that
$[(p \vee q) \wedge (\sim(\sim p \wedge \sim q \vee r))] \vee (\sim p \wedge q) \vee (\sim p \wedge \sim$
is a ~~tautology~~ contigency.

* Logical Equivalency
Two statements A & B are said to be logically equivalent if the truth value of A and truth values of B are same.

Show that following statements are logical equivalent :
1) $p \equiv (p \leftrightarrow q) \equiv (\sim p \vee q) \wedge (\sim q \vee p)$

p	q	$p \leftrightarrow q$	$\sim p$	$\sim q$	$\sim p \vee q$	$\sim q \vee p$	$(\sim p \vee q) \wedge (\sim q$
T	T	T	F	F	T	T	T
T	F	F	F	T	F	T	F
F	T	F	T	F	T	F	F
F	F	T	T	T	T	T	T

$(p \leftrightarrow q) \equiv (\sim p \vee q) \wedge (\sim q \vee p)$

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2) $p \to (q \wedge r) \equiv (p \to q) \wedge (p \to r)$

p	q	r	$(q \wedge r)$	$p \to (q \wedge r)$
T	T	T	T	T
T	T	F	F	F
T	F	T	F	F
T	F	F	F	F
F	T	T	T	T
F	T	F	F	T
F	F	T	F	T
F	F	F	F	T

$p \to q$	$p \to r$	$(p \to q) \wedge (p \to r)$
T	T	T
T	F	F
F	T	F
F	F	F
T	T	T
T	T	T
T	T	T
T	T	T

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Que: State and prove De-Morgan's law.

Ans: The negation of conjunction of two or more ~~stat~~ statements is logically equivalent to the disjunction of negation of the statements

If p and q are two statements then

$\sim(p \wedge q) \equiv \sim p \vee \sim q$

p	q	$p \wedge q$	$\sim(p \wedge q)$	$\sim p$	$\sim q$	$\sim p \vee \sim q$
T	T	T	F	F	F	F
T	F	F	T	F	T	T
F	T	F	T	T	F	T
F	F	F	T	T	T	T

The negation of disjunction of two or more statement is logically equivalent to conjunction of negation of the statements.

$\sim(p \vee q) \equiv \sim p \wedge \sim q$

p	q	$p \vee q$	$\sim(p \vee q)$	$\sim p$	$\sim q$	$\sim p \wedge \sim q$
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F
F	F	F	T	T	T	T

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* Argument & Validity

An argument is an assert that a given set of prepositions $S_2, S_3 \ldots S_n$ is valid if and on $S_1 \wedge S_2 \wedge S_3 \ldots S_n \vdash S$ is a tautol

Ex: S_1 : If sachin plays ~~ctr~~ cricket. we will win the Match

S_2 : If we win the match we be happy

$\Rightarrow$ p : Sachin plays a cricket

q : we will win the the match

r : We will be happy

Check

~~Prove~~ that the following argument valid. $p \to \sim q$; $r \to q$, r $\therefore \sim p$

Sol^n: S_1 : $p \to \sim q$

S_2 : $r \to q$

S_3 : r

S : $\sim p$

To prove : $S_1 \wedge S_2 \wedge S_3 \to S$

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		S_3	S		S_1	S_2
p	q	r	$\sim p$	$\sim q$	$p \to \sim q$	$r \to q$
T	T	T	F	F	F	T
T	T	F	F	F	F	T
T	F	T	F	T	T	F
T	F	F	F	T	T	T
F	T	T	T	F	T	T
F	T	F	T	F	T	T
F	F	T	T	T	T	F
F	F	F	T	T	T	T

$S_1 \wedge S_2$	$S_1 \wedge S_2 \wedge S_3$	$S_1 \wedge S_2 \wedge S_3 \to S$
F	F	T
F	F	T
F	F	T
T	F	T
T	T	T
T	F	T
F	F	T
T	F	T

Hence it is Tautology

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Que: Determine the validity of following argu

If it rains, the street gets wet

If it rains, accident happens

The street gets wet

∴ Accident happen

Sol^n: Let p : It rains

q : The street get wet

r : Accident happens

∴ $S_1 : p \rightarrow q$

$S_2 : p \rightarrow r$

$S_3 : q$

$S : r$

The argument will be valid if $(S_1 \wedge S_2 \wedge$ is a tautology.

$[((p \rightarrow q) \wedge (p \rightarrow r) \wedge q) \rightarrow r]$ is a tau

p	q	r	$p \rightarrow q$	$p \rightarrow r$	$(p \rightarrow q) \wedge (p \rightarrow r) \wedge q$		—||—
T	T	T	T	T	T		T
T	T	F	T	F	F		T
T	F	T	F	T	F		T
T	F	F	F	F	F		T
F	T	T	T	T	F		T
F	T	F	T	T	T		T
F	F	T	T	T	T		F
F	F	F	T	T	F		T

$[((p \rightarrow q) \wedge (p \rightarrow r) \wedge q) \rightarrow r]$ is not a tautology

∴ Statement is invalid.

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Que: Check the validity of the following argume

If it rains today then we will not have a party today.

If we don't have a party today then we will have a party tomorrow.

∴ If it rains today, then we will have a party tomorrow.

Sol^n: Let p : It rains today

q : We will have a party today

r : We will have a party tomorrow.

∴ $S_1 : p \rightarrow \sim q$

$S_2 : \sim q \rightarrow r$

$S : p \rightarrow r$

The argument will be valid if $(S_1 \wedge S_2) \rightarrow S$

$[(p \rightarrow \sim q) \wedge (\sim q \rightarrow r)] \rightarrow (p \rightarrow r)$

p	q	r	$\sim q$	$p \rightarrow \sim q$ (1)	$\sim q \rightarrow r$ (2)	(1) ∧ (2) (4)	$p \rightarrow r$ (3)	(4) → (3)
T	T	T	F	F	T	F	T	T
T	T	F	F	F	T	F	F	T
T	F	T	T	T	T	T	T	T
T	F	F	T	T	F	F	F	T
F	T	T	F	T	T	T	T	T
F	T	F	F	T	T	T	T	T
F	F	T	T	T	T	T	T	T
T	F	F	T	T	F	F	F	T

$[(p \rightarrow \sim q) \wedge (\sim q \rightarrow r)] \rightarrow (p \rightarrow r)$ is a tautology

∴ Statement is ~~tau~~ valid.

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* Basic Concept of set theory

A set is defined as an unordered collection of well-defined objects. The objects are called elements of the set. There are two different of writing a set.

① Listing method

In this method all elements of set are separated by ',' & the elements are unique. It is also known as ?

Ex: $A = \{1, 2, 3, 4, 5\}$

② Set Builder Method

In this method all the elements of the set are denoted by 'x' where satisfies certain property.

Ex: $A = \{x \in N \mid x \leq 5\}$

Types of Sets

1] Finite set: A set A is said to be finite if it contains finite no. of elements

Ex: $A = \{1, 2, 3, 4, 5\}$

2] Infinite set: A set A is said to be infinite if it contains infinite no. of elements

Ex: $A = \{$

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3] Singleton set: A set A is said to be singleton set if it contains only 1 element

Ex: $A = \{5\}$

4] Null / empty set: A set A is said to be null set or empty set if it contains no elements. It is denoted by ϕ

Ex: $A = \{\ \} = \phi$

5] Subset: A set A is said to be subset of set B if and only if every element of A is also an element of B.

Ex: $A \subseteq B$.

6] Proper Subset: A set A is called a proper subset of set B if:

(i) A is subset of B

(ii) B is not a subset of A

Ex: $A = \{x, y, z\}$

$B = \{x, y, z, w, a\}$

$A \subset B$

7] Equal Sets: Two non empty sets A & B are said to be equal if:- $A \subseteq B$, $B \subseteq A$

8] Universal set: All the sets are likely to be consider as subset of a particular set and it is denoted by U.